Technical University of Crete

School of Electrical and Computer Engineering

Course: **Optimization**

Exercise 1 (80/1000)

Report Delivery Date: October 28, 2025

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In this exercise, we will study simple concepts from Calculus of Several Variables (Taylor expansions), Convex Sets, and Convex Functions (Note: $\mathbb{R}_+ = \{x \in \mathbb{R} \,|\, x \geq 0\}$, $\mathbb{R}_{++} = \{x \in \mathbb{R} \,|\, x > 0\}$).

`You must prepare an electronic (LaTeX) report`
`and upload the pdf file and the related code.`

1. (10) Let $f : \mathbb{R}_+ \to \mathbb{R}$, with $f(x) = \frac{1}{1+x}$. Let $x_0 \in \mathbb{R}_+$, and define the first- and second-order Taylor approximations of f at x_0 as

$$\begin{aligned} f_{(1)}(x) &= f(x_0) + f'(x_0)(x - x_0), \\ f_{(2)}(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2. \end{aligned} \tag{1}$$

   (a) (5) Find the analytic expressions for functions f' and f'';

   (b) (5) Draw in a common plot $f(x)$, $f_{(1)}(x)$ and $f_{(2)}(x)$ and, in order to understand the behavior of the approximations, consider various values of x_0.

2. (20) Let $f : \mathbb{R}_+^2 \to \mathbb{R}$, with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$.

   (a) (0) Compute and plot, via `mesh`, f for $x_1, x_2 \in [0, x_*]$, with $x_* > 0$.

   (b) (5) Plot the level sets of f, via `contour`. What do you observe? Can you explain the phenomenon?

   (c) (5) Compute the first- and second-order Taylor approximations of f at point $\mathbf{x}_0 = (x_{0,1}, x_{0,2})$.

   (d) (5) Draw on a common plot, via `mesh`, f and its first-order Taylor approximation.

   (e) (5) Draw on a common plot, via `mesh`, f and its second-order Taylor approximation.

3. (20) Let $\mathbb{S}_{\mathbf{a},b} = \{\mathbf{x} \in \mathbb{R}^n \,|\, \mathbf{a}^T\mathbf{x} \leq b\}$.

   (a) (5) Prove that $\mathbb{S}_{\mathbf{a},b}$ is convex.

   (b) (15) Prove that $\mathbb{S}_{\mathbf{a},b}$ is *not* affine (a counterexample is sufficient).

4. (10) Find the point $\mathbf{x}_*$ that is co-linear with $\mathbf{a}$ and lies on the hyperplane $\mathbb{H}_{\mathbf{a},b} := \{\mathbf{x} \in \mathbb{R}^n \,|\, \mathbf{a}^T\mathbf{x} = b\}$.

5. (20) Check whether the following functions are convex or not (using, for example, the second derivative rule).

   (a) (5) $f : \mathbb{R}_+ \to \mathbb{R}$, with $f(x) = \frac{1}{1+x}$;

   (b) (5) $f : \mathbb{R}_+^2 \to \mathbb{R}$, with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$;

   (c) (5) $f : \mathbb{R}_{++} \to \mathbb{R}$, with $f(x) = x^a$, for (to get a better feeling, plot function x^a, for various values of a)

      i. $a \geq 1$ and $a \leq 0$;

      ii. $0 \leq a \leq 1$.

   (d) (5) $f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$, with $f_1(\mathbf{x}) = \|\mathbf{x}\|_2$ and $f_2(\mathbf{x}) = \|\mathbf{x}\|_2^2$ (plot the functions for $n = 2$).

6. (10) In the sequel, we shall study the very important convex quadratic functions. Let $\mathbf{P} \in \mathbb{S}_{++}^2$, where $\mathbb{S}_{++}^2$ is the set of the (2×2) positive definite matrices, $\mathbf{q} \in \mathbb{R}^2$, and $r \in \mathbb{R}$.[1] Consider the convex quadratic function $f : \mathbb{R}^2 \to \mathbb{R}$, defined as:

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{P}\mathbf{x} + \mathbf{q}^T\mathbf{x} + r. \tag{3}$$

   It can be shown (we shall prove it later) that the function f is minimized at point

$$\mathbf{x}_* = -\mathbf{P}^{-1}\mathbf{q}. \tag{4}$$

   (a) (10) Plot function f, via `mesh` and `contour` around $\mathbf{x}_*$, and verify, visually, the claim about the optimal point $\mathbf{x}_*$.

[1] In order to generate such a $\mathbf{P}$, do as follows:

$$\begin{aligned} \mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}} &= \texttt{randn}(2,2); \\ \mathbf{P} &= \mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}} * \mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}}'; \end{aligned} \tag{2}$$

The generated matrix $\mathbf{P}$ is positive definite with probability 1.

(b) (20) Let $\mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $f : \mathbb{R}^n \to \mathbb{R}$, with $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$.

i. (10) Assume that the columns of $\mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}}$ are linearly independent and prove that f is strictly convex (Prove that the Hessian of $f(\mathbf{x})$ is positive definite).

ii. (10) Plot f for $m = 3$ and $n = 2$. In order to generate the data, generate a random (3×2) matrix $\mathbf{1} + \mathbf{x_{0,1}} + \mathbf{x_{0,2}}$, a random (2×1) vector $\mathbf{x}$, and compute $\mathbf{b} = \mathbf{A}\mathbf{x}$. Then, plot, via `mesh`, function f in a square around the true value $\mathbf{x}$ - use also the `contour` statement. What do you observe?

Repeat the above procedure by assuming that $\mathbf{b} = \mathbf{A}\mathbf{x} + \mathbf{e}$, where $\mathbf{e}$ is a "small noise" vector. What do you observe?

SOLUTIONS

1. (a) First Derivative:

$$f'(x) = -\frac{1}{(1+x)^2}$$

Second Derivative:

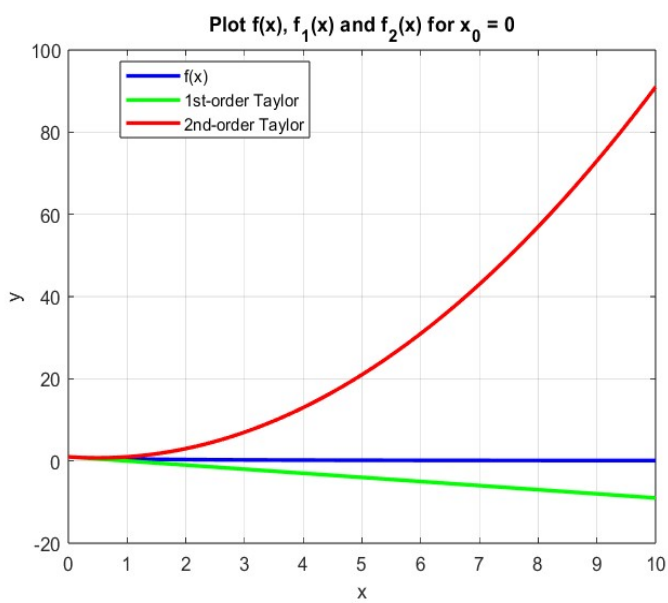
$$f''(x) = \frac{2}{(1+x)^3}$$

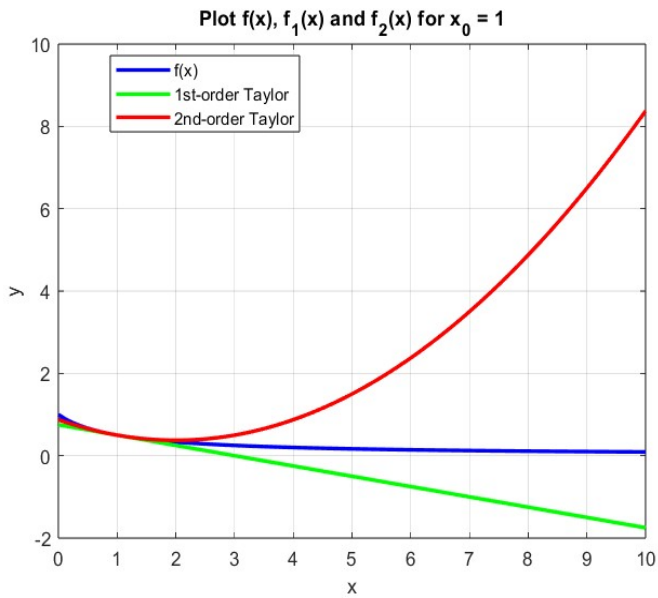
$$f_{(1)}(x) = \frac{1}{(1+x_0)} - \frac{1}{(1+x_0)^2}(x - x_0)$$

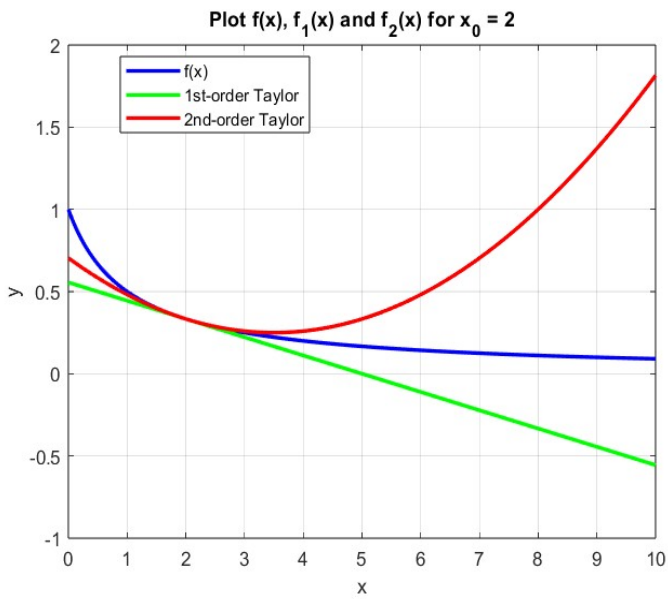
$$f_{(2)}(x) = \frac{1}{(1+x_0)} - \frac{1}{(1+x_0)^2}(x - x_0) + \frac{1}{2}\frac{2}{(1+x_0)^3}(x - x_0)^2$$

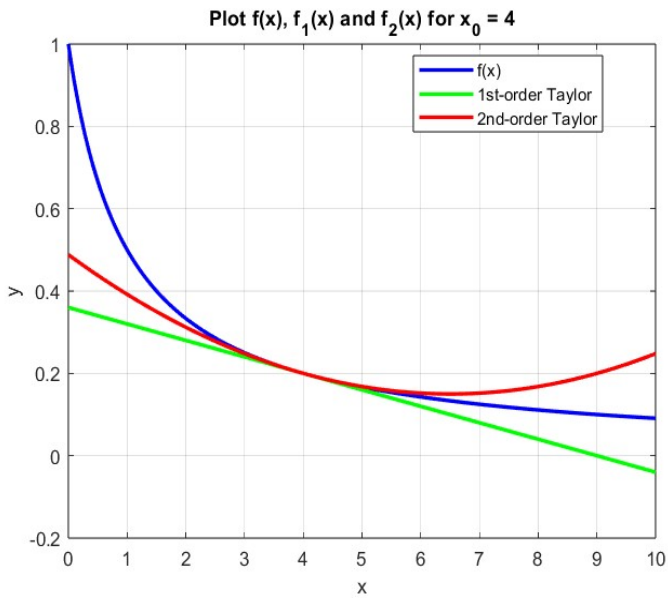
$$= \frac{1}{(1+x_0)} - \frac{1}{(1+x_0)^2}(x - x_0) + \frac{1}{(1+x_0)^3}(x - x_0)^2$$

(b)

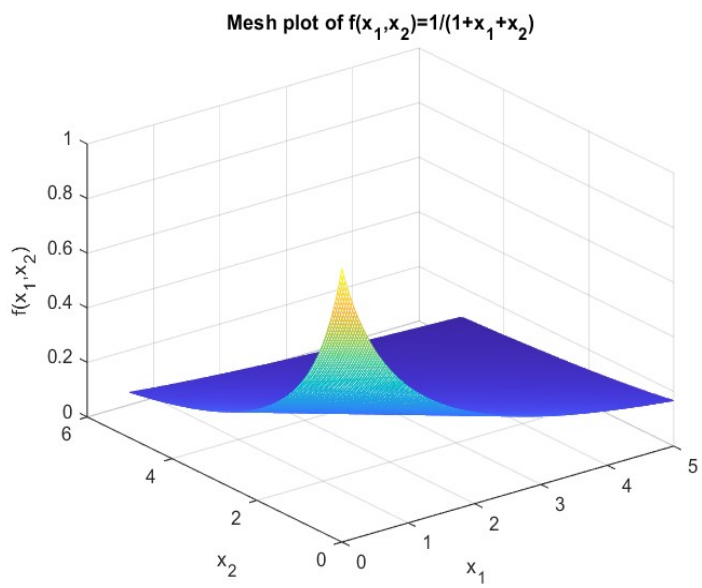



Plot f(x), f1(x) and f2(x) for x0 = 1
f(x)
1st-order Taylor
2nd-order Taylor
y
x

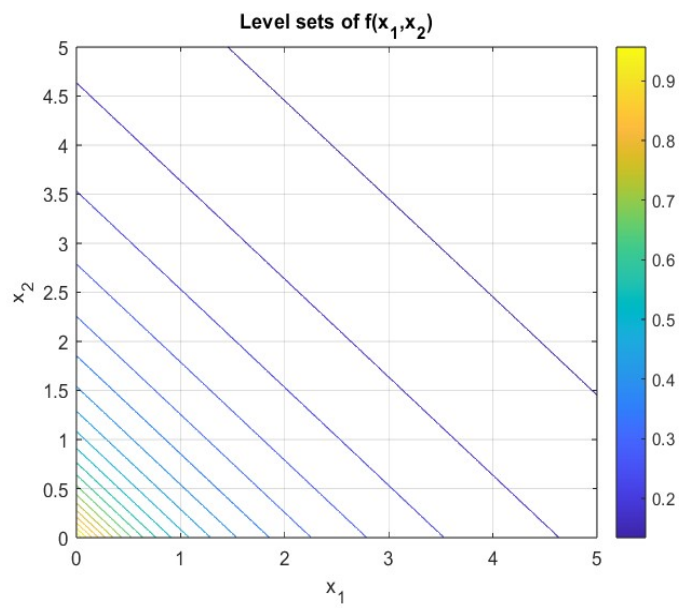

Plot f(x), f1(x) and f2(x) for x0 = 2
f(x)
1st-order Taylor
2nd-order Taylor
y
x


Plot f(x), f1(x) and f2(x) for x0 = 4
f(x)
1st-order Taylor
2nd-order Taylor
y
x
1
0.8
0.6
0.4
0.2
0
-0.2
0
1
2
3
4
5
6
7
8
9
10

2. (a)



(b)



Let $f(x_1, x_2) = c$.

$$\frac{1}{1+x_1+x_2} = c \implies 1 + x_1 + x_2 = \frac{1}{c} \implies x_1 + x_2 = \frac{1}{c} - 1$$

$$\implies x_2 + x_1 = \frac{1}{c} - 1 = C \implies x_2 = C - x_1$$

The level sets are parallel straight lines with slope -1.

This happens because the function depends only on the sum $x_1 + x_2$. For any constant value of $x_1 + x_2$, the function takes the same value, which results in level sets that are straight lines with slope -1.

(c)

$$f(x_1, x_2) = \frac{1}{1 + x_1 + x_2}$$

First partial Derivatives:

$$\frac{\partial f}{\partial x_1} = -1 \cdot (1 + x_1 + x_2)^{-2} \cdot 1 = -(1 + x_1 + x_2)^{-2}$$

$$\frac{\partial f}{\partial x_2} = -1 \cdot (1 + x_1 + x_2)^{-2} \cdot 1 = -(1 + x_1 + x_2)^{-2}$$

Second partial Derivatives:

$$\frac{\partial^2 f}{\partial x_1^2} = -(-2)(1 + x_1 + x_2)^{-3} \cdot 1 = 2(1 + x_1 + x_2)^{-3}$$

$$\frac{\partial^2 f}{\partial x_2^2} = -(-2)(1 + x_1 + x_2)^{-3} \cdot 1 = 2(1 + x_1 + x_2)^{-3}$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = -(-2)(1 + x_1 + x_2)^{-3} \cdot 1 = 2(1 + x_1 + x_2)^{-3}$$

$$H_f(\mathbf{x}_0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}_0) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}_0) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}_0) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}_0) \end{pmatrix} = \begin{pmatrix} \frac{2}{(1+x_{0,1}+x_{0,2})^3} & \frac{2}{(1+x_{0,1}+x_{0,2})^3} \\ \frac{2}{(1+x_{0,1}+x_{0,2})^3} & \frac{2}{(1+x_{0,1}+x_{0,2})^3} \end{pmatrix}$$

First order Taylor approximations of f:

$$f_{(1)}(x_1, x_2) = f(x_{0,1}, x_{0,2}) + \frac{\partial f}{\partial x_1}(x_0)(x_1 - x_{0,1}) + \frac{\partial f}{\partial x_2}(x_0)(x_2 - x_{0,2})$$

$$= \frac{1}{1 + x_1 + x_2} - \frac{1}{(1 + x_1 + x_2)^2}(x_1 - x_{0,1}) - \frac{1}{(1 + x_1 + x_2)^2}(x_2 - x_{0,2})$$

Second order Taylor approximations of f:

$$f_{(2)}(x_1, x_2) = f_{(1)}(x_1, x_2) + \frac{1}{2}\frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}_0)(x_1 - x_{0,1})^2 + 2\frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}_0)(x_1 - x_{0,1})(x_2 - x_{0,2}) +$$

$$\frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}_0)(x_2 - x_{0,2})^2$$

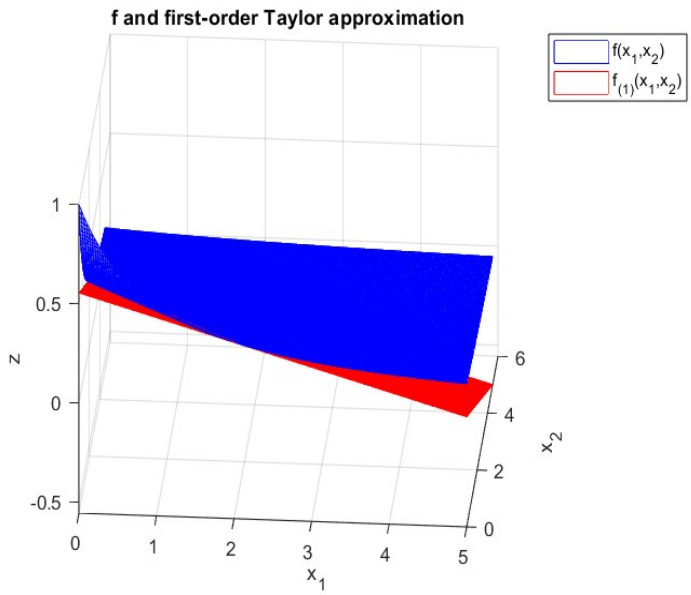
$$= f_{(1)}(x_1, x_2) + \frac{1}{2}\frac{2}{1 + x_{0,1} + x_{0,2}^3}(x_1 - x_{0,1})^2 + 2\frac{2}{1 + x_{0,1} + x_{0,2}^3}(x_1 - x_{0,1})(x_2 - x_{0,2}) +$$

$$\frac{2}{1+x_{0,1}+x_{0,2}^3}(x_2-x_{0,2})^2$$

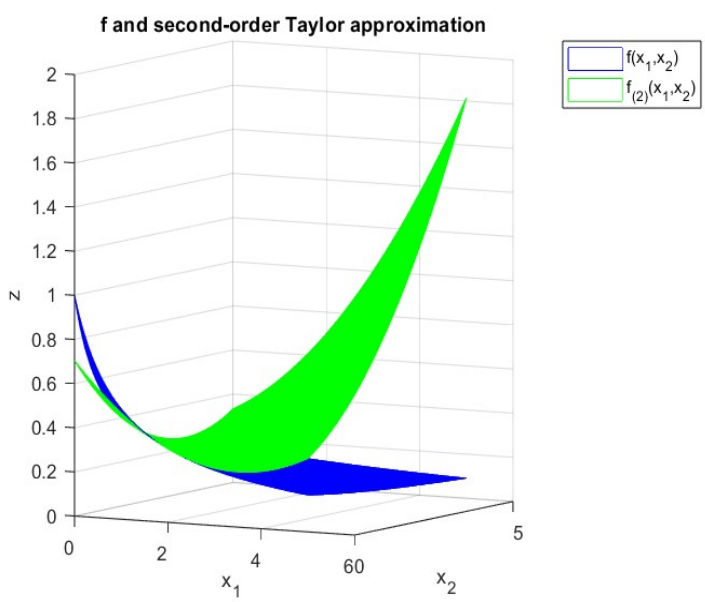
$$= f_{(1)}(x_1,x_2)+\frac{1}{1+x_{0,1}+x_{0,2}^3}(x_1-x_{0,1})^2+2(x_1-x_{0,1})(x_2-x_{0,2})+(x_2-x_{0,2})^2$$

$$= f_{(1)}(x_1,x_2)+\frac{1}{1+x_{0,1}+x_{0,2}^3}(x_1-x_{0,1})+(x_2-x_{0,2})^2$$

(d)



(e)



(a) i. Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be two arbitrary points in $S_{\mathbf{a},b}$.

By the definition of $S_{\mathbf{a},b}$, this means:

$$\mathbf{a}^T\mathbf{x}_1 \leq b \quad \text{and} \quad \mathbf{a}^T\mathbf{x}_2 \leq b$$

Let λ be any scalar such that $\lambda \in [0, 1]$.

Consider the convex combination $\mathbf{x}_\lambda = \lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$.

We must show that $\mathbf{x}_\lambda \in S_{\mathbf{a},b}$, i.e., that $\mathbf{a}^T\mathbf{x}_\lambda \leq b$.

Substitute $\mathbf{x}_\lambda$ into the inequality:

$$\mathbf{a}^T\mathbf{x}_\lambda = \mathbf{a}^T(\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) = \lambda(\mathbf{a}^T\mathbf{x}_1) + (1-\lambda)(\mathbf{a}^T\mathbf{x}_2)$$

Use the initial inequalities and the constraint on λ:Since $\mathbf{x}_1, \mathbf{x}_2 \in S_{\mathbf{a},b}$, we have $\mathbf{a}^T\mathbf{x}_1 \leq b$ and $\mathbf{a}^T\mathbf{x}_2 \leq b$. Since $\lambda \in [0, 1]$, we have $\lambda \geq 0$ and $(1-\lambda) \geq 0$. Multiplying the inequalities by the non-negative scalars λ and $(1-\lambda)$:

$$\lambda(\mathbf{a}^T\mathbf{x}_1) \leq \lambda b$$

$$(1-\lambda)(\mathbf{a}^T\mathbf{x}_2) \leq (1-\lambda)b$$

Add the two inequalities:

$$\lambda(\mathbf{a}^T\mathbf{x}_1) + (1-\lambda)(\mathbf{a}^T\mathbf{x}_2) \leq \lambda b + (1-\lambda)b$$

$$\mathbf{a}^T\mathbf{x}_\lambda \leq (\lambda + 1 - \lambda)b$$

$$\mathbf{a}^T\mathbf{x}_\lambda \leq b$$

Since the inequality holds, $\mathbf{x}_\lambda \in S_{\mathbf{a},b}$.

Therefore, $S_{\mathbf{a},b}$ is convex.

ii. For $S_{\mathbf{a},b}$ to be affine, it must satisfy $\mathbf{a}^T(\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \leq b$ for all $\lambda \in \mathbb{R}$.

The set is $S_{\mathbf{a},b} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T\mathbf{x} \leq b\}$.

Let $n = 1$, with $\mathbf{a} = 1$ and $b = 0$. The set now is: $S_{1,0} = \{x \in \mathbb{R} \mid x \leq 0\} = (-\infty, 0]$

We choose two points in the set: $x_1 = -1$ and $x_2 = 0$.

Both satisfy the condition $x \leq 0$, so $x_1, x_2 \in S_{1,0}$.

Let $\lambda = -\mathbf{1}$. Calculate the affine combination:

$$\mathbf{x}_\lambda = \lambda x_1 + (1-\lambda)x_2 = (-1)(-1) + (1-(-1))(0) = 1 + (2)(0) = 1$$

The point $x_\lambda = 1$ is in $S_{1,0}$ if $1 \leq 0$. This is incorrect.

Since $x_1, x_2 \in S_{1,0}$ but their affine combination $x_\lambda = 1$ is not in $S_{1,0}$, the set $S_{\mathbf{a},b}$ is not affine.

(b) If $\mathbf{x}_*$ is collinear with $\mathbf{a}$, it means $\mathbf{x}_*$ must be a scalar multiple of $\mathbf{a}$.

$$\mathbf{x}_* = \lambda \mathbf{a} \tag{5}$$

for some scalar $\lambda \in \mathbb{R}$.

Now, substitute this expression for $\mathbf{x}_*$ into the equation of the hyperplane $H_{\mathbf{a},b}$:

$$\mathbf{a}^T \mathbf{x}_* = b$$

Substitute $\mathbf{x}_* = \lambda \mathbf{a}$:

$$\mathbf{a}^T (\lambda \mathbf{a}) = b$$

Now, we solve for λ:

$$\lambda (\mathbf{a}^T \mathbf{a}) = b$$

Recall that $\mathbf{a}^T \mathbf{a} = \|\mathbf{a}\|_2^2$.

$$\lambda \|\mathbf{a}\|_2^2 = b$$

To solve for λ, we assume $\mathbf{a}$ is a non-zero vector.

$$\lambda = \frac{b}{\|\mathbf{a}\|_2^2}$$

So from (5):

$$\mathbf{x}_* = \lambda \mathbf{a} = \left(\frac{b}{\|\mathbf{a}\|_2^2} \right) \mathbf{a}$$

(c) i. $f : \mathbb{R}_+ \to \mathbb{R}$, with $f(x) = \frac{1}{1+x}$.

The domain is $Dom(f) = \mathbb{R}_+ \geq 0$, which is a convex set.

First Derivative: $f'(x) = -\frac{1}{(1+x)^2}$

First Derivative: $f''(x) = \frac{2}{(1+x)^3}$

Check for Convexity: For all $x \in \mathbb{R}_+$, we have $1 + x \geq 0$, so $(1 + x)^3 \geq 0$.

Therefore, $f''(x) = \frac{2}{(1+x)^3} \geq 0$

Since the second derivative is strictly positive on the entire domain, the function f(x) is convex.

ii. $f : \mathbb{R}^2_+ \to \mathbb{R}$, with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$.

The domain is $Dom(f) = \mathbb{R}^2_+ = (x_1, x_2)|x_1 \geq 0, x_2 \geq 0$, which is a convex set.

Second Partial Derivatives:

$\frac{\partial^2 f}{\partial x_1^2} = \frac{2}{(1+x_1+x_2)^3}$

$\frac{\partial^2 f}{\partial x_2^2} = \frac{2}{(1+x_1+x_2)^3}$

$f_{x_1x_2} = \frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{2}{(1+x_1+x_2)^3} = f_{x_2x_1}$

Hessian Matrix:

Let $C = \frac{2}{(1+x_1+x_2)^3}$. Since $x_1, x_2 \geq 0$, C is a positive constant ($C > 0$).

$H = \begin{bmatrix} f_{x_1x_1} & f_{x_1x_2} \\ f_{x_2x_1} & f_{x_2x_2} \end{bmatrix} = \begin{bmatrix} \mathrm{C} & \mathrm{C} \\ \mathrm{C} & \mathrm{C} \end{bmatrix} = 2(1 + x_1 + x_2)^{-3} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Check for Convexity: For the function to be convex, the Hessian must be positive semidefinite.

This means the eigenvalues must be ≥ 0, or that the principal minors must satisfy the appropriate conditions.

Leading Principal Minor 1: $\det(H) = C$. Since $C > 0$, the first minor is positive.

Leading Principal Minor 2 (Determinant of H):

$$\det(H) = C \cdot C - C \cdot C = C^2 - C^2 = 0$$

Since $\det(\nabla^2 f(\mathbf{x})) = 0$, the Hessian is positive semidefinite.

Therefore, the function $f(x_1, x_2)$ is convex.

iii. $f : \mathbb{R}_{++} \to \mathbb{R}$, with $f(x) = x^a$.

The domain is $\mathrm{Dom}(f) = \mathbb{R}_{++} = (0, \infty)$, which is a convex set.

First Derivative:

$$f'(x) = a * x^{a-1}$$

Second Derivative:

$$f''(x) = a(a-1) * x^{a-2}$$

The convexity depends entirely on the sign of $f''(x)$.

Since $x \in (0, \infty)$, we have $x^{a-2} > 0$.

The sign of $f''(x)$ is determined by the sign of the constant term $a(a-1)$

I: $a \geq 1$ and $a \leq 0$

Case $a \geq 1$:

If $a \geq 1$, then $a > 0$ and $(a-1) \geq 0$.

Thus, $a(a-1) \geq 0$. This means $f''(x) = a(a-1)x^{a-2} \geq 0$.

The function $f(x)$ is convex when $a \geq 1$.

Case $a \leq 0$:

If $a \leq 0$, then $a \leq 0$ and $(a-1) < 0$.

Thus, $a(a-1) \geq 0$ (a non-positive times a negative is non-negative).This means $f''(x) = a(a-1)x^{a-2} \geq 0$.

The function $f(x) = x^a$ is convex when $a \leq 0$.

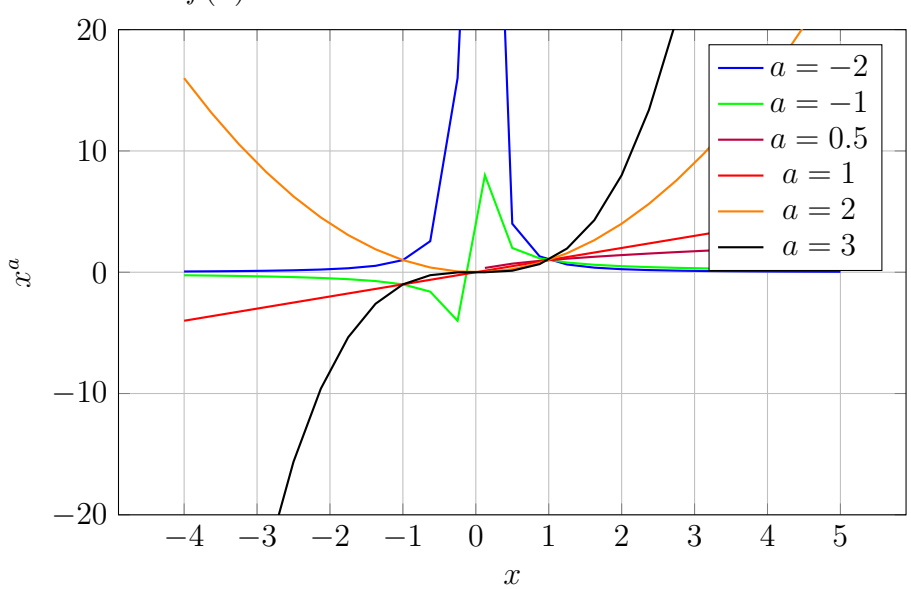
II. $0 < a < 1$

Case $0 < a < 1$:

If $0 < a < 1$, then $a > 0$ and $(a-1) < 0$.

Thus, $a(a-1) < 0$.This means $f''(x) = a(a-1)x^{a-2} < 0$.

The function $f(x) = x^a$ is concave when $0 < a < 1$.



iv. The domain for both functions is $\mathbb{R}^n$, which is a convex set.

$f_1(\mathbf{x}) = \|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

Proof using the definition of convexity (Triangle Inequality):

We need to show $f_1(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \leq \lambda f_1(\mathbf{x}) + (1-\lambda) f_1(\mathbf{y})$ for $\lambda \in [0, 1]$.

$$f_1(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) = \|\lambda\mathbf{x} + (1-\lambda)\mathbf{y}\|_2$$

Using the Triangle Inequality for norms:

$$\|\lambda\mathbf{x} + (1-\lambda)\mathbf{y}\|_2 \leq \|\lambda\mathbf{x}\|_2 + \|(1-\lambda)\mathbf{y}\|_2$$

Using the property of norms (homogeneity: $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$):

$$\|\lambda\mathbf{x}\|_2 + \|(1-\lambda)\mathbf{y}\|_2 = |\lambda|\|\mathbf{x}\|_2 + |1-\lambda|\|\mathbf{y}\|_2$$

Since $\lambda \in [0,1]$, we have $|\lambda| = \lambda$ and $|1-\lambda| = 1-\lambda$.

$$\|\lambda\mathbf{x}\|_2 + \|(1-\lambda)\mathbf{y}\|_2 = \lambda\|\mathbf{x}\|_2 + (1-\lambda)\|\mathbf{y}\|_2 = \lambda f_1(\mathbf{x}) + (1-\lambda)f_1(\mathbf{y})$$

Thus, $f_1(\mathbf{x}) = \|\mathbf{x}\|_2$ is convex.

$f_2(\mathbf{x}) = \|\mathbf{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$
This is the squared Euclidean norm.
We will use the Hessian matrix.
Partial Derivatives:

$$\frac{\partial f_2}{\partial x_i} = 2x_i$$

Second Partial Derivatives:

$$\frac{\partial^2 f_2}{\partial x_i^2} = 2$$

$$\frac{\partial^2 f_2}{\partial x_i \partial x_j} = 0 \quad \text{for } i \neq j$$

Hessian Matrix:

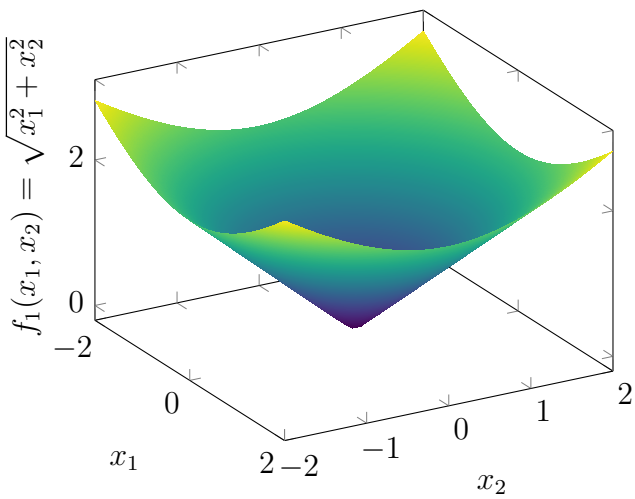
$$H = \begin{pmatrix} 2 & 0 & \cdots & 0 \\ 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 2 \end{pmatrix} = 2\mathbf{I}$$

Check for Convexity (Positive Semidefiniteness):
For any vector $\mathbf{v} \in \mathbb{R}^n$, the quadratic form is:

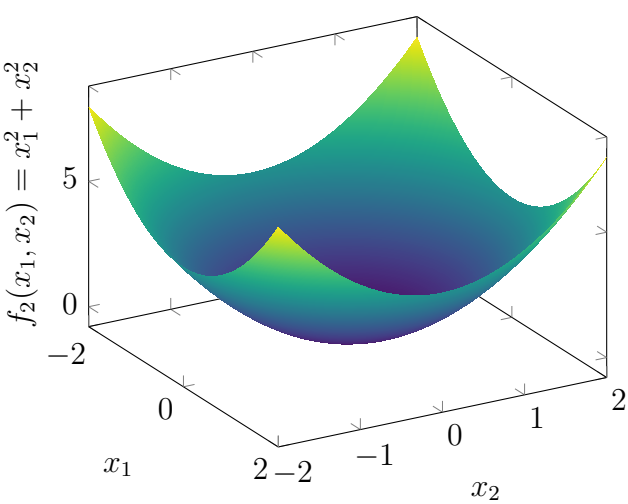
$$\mathbf{v}^T \nabla^2 f_2(\mathbf{x}) \mathbf{v} = \mathbf{v}^T (2\mathbf{I}) \mathbf{v} = 2\mathbf{v}^T \mathbf{v} = 2\|\mathbf{v}\|_2^2$$

Since $2\|\mathbf{v}\|_2^2 > 0$ for all $\mathbf{v} \neq \mathbf{0}$, the Hessian is positive definite, and thus positive semidefinite.
Therefore, the function $f_2(\mathbf{x}) = \|\mathbf{x}\|_2^2$ is convex.

$f_1(x_1, x_2) = \|\mathbf{x}\|_2$



$f_2(x_1, x_2) = \|\mathbf{x}\|_2^2$



(d)